GROUP PROJECT REPORT

Marketing Mix Modelling & Budget Optimization

Measuring the sales impact of six advertising channels and finding the best split of a fixed budget for a Sri Lankan FMCG brand

Dataset	Monthly marketing data, January 2023 – December 2025 (36 months)
Tools	Python (Pandas, NumPy, Scikit-learn, SciPy, Statsmodels, Matplotlib, Seaborn), Power BI
Team	10 members · Team lead: Indumini Liyanage
Repository	github.com/IBL11/mmm-budget-optimization
Submission date	13 November 2026

Contents

Executive Summary 3

1. Introduction and Objectives 4
- 1.1 Objectives 4
- 1.2 Deliverables 4

2. Data 5
- 2.1 Cleaning and Feature Engineering 5
- 2.2 Exploratory Data Analysis 5

3. Methodology 6

4. Model Results and Validation 7
- 4.1 Test Performance 7
- 4.2 Validation Checks 8

5. Channel Effectiveness and ROI 9
- 5.1 Base Sales vs Advertising 9
- 5.2 ROI by Channel 9
- 5.3 Diminishing Returns 10
- 5.4 External Factors 11

6. Budget Optimization 12
- 6.1 Set-up 12
- 6.2 Recommended Split 12
- 6.3 Budget Scenarios 13
- 6.4 Sensitivity 13

7. Interactive Dashboard 14

8. Recommendations, Limitations and Conclusion 15
- 8.1 Recommendations 15
- 8.2 Limitations 15
- 8.3 Future Work 15
- 8.4 Conclusion 15

Appendix: Repository and Reproduction 16

Executive Summary

Expected sales uplift	Model test error (MAPE)	Return on ad spend	Best channels
+LKR 25.9M	**1.13%**	**LKR 2.31**	**YouTube & Google**
+2.5% with the same budget	R^2 0.938 on 6 unseen months	Sales per LKR 1 of advertising	Marginal ROI 2.13 and 1.92

Re-splitting the same **LKR 240.3M** annual advertising budget, moving money from Facebook/Instagram and Radio to Google Search and YouTube, is predicted to raise 2025 sales by **LKR 25.9M (+2.5%)** at no extra cost.

We built a Marketing Mix Model (MMM) on 36 months of Sri Lankan FMCG data. The model captures delayed advertising effects (adstock), diminishing returns (Hill saturation) and seven external drivers such as promotions, seasonality and competitor pressure. It predicts the six held-out test months with **R^2 0.938** and **1.13% average error (MAPE)**.

- **About half of sales (49.6%) would happen without advertising.** Advertising drives the other 50.4% and returns about LKR 2.31 in sales per LKR 1 spent.
- **YouTube (ROI 4.33) and Google Search (ROI 3.93) are the most effective channels** and are under-funded relative to the sales they drive.
- **Facebook/Instagram is over-funded:** 24% of spend but 10% of advertising-driven sales, with a marginal ROI of 0.50 (below break-even).
- **Radio shows no measurable effect** in this data and should be tested before further cuts.
- **Promotions and seasonality** move monthly sales more than any single media channel.
- **The findings are robust.** All residual tests pass, multicollinearity was fixed (maximum VIF 6.3), and the channel ranking holds under a bootstrap, an alternative model, nine adstock/saturation settings and different budget limits.

> **Recommendation:** adopt the optimized split gradually (within ±30% per channel per year), improve the split before increasing the total budget, and re-run the model every year.

1. Introduction and Objectives

The project answers one business question: **how should a fixed advertising budget be split across channels to get the most sales?** Total sales are recorded once a month, so the effect of each channel cannot be read from the data directly. A Marketing Mix Model separates it.

1.1 Objectives

1. Measure how much each advertising channel adds to sales, after removing external factors.
2. Capture delayed effects (adstock) and diminishing returns (saturation).
3. Calculate each channel's ROI and marginal ROI.
4. Find the budget split that maximises predicted sales under a fixed budget.
5. Present the results in an interactive Power BI dashboard.

1.2 Deliverables

Table 1. *Project deliverables*

Deliverable	Location in the repository
Cleaned data and EDA	notebooks 01–03, data_cleaning_report.md, eda_summary.md
Marketing Mix Model and validation	notebooks 04–05, model_evaluation_and_handoff.md
Channel effectiveness and ROI	notebook 06, roi_analysis_report.md
Budget optimization algorithm	notebook 07, budget_optimization_report.md
Interactive dashboard	dashboard/MMM_Marketing_Dashboard.pbix
Final report	This document

2. Data

The dataset is complete and consistent: 36 consecutive months, no missing values and no duplicates. It is synthetic data for a Sri Lankan FMCG packaged-foods brand, provided for analytical practice.

Table 3. *Dataset overview*

Item	Detail
Period	January 2023 – December 2025, monthly (36 rows)
Target	sales_revenue_lkr (LKR)
Media channels (6)	TV, Facebook/Instagram, Google Search, YouTube, Radio, Outdoor
External factors (8)	Price index, promotion discount, competitor pressure, economic sentiment, LKR/USD exchange rate, inflation proxy, seasonal demand index, public holidays
Total ad spend	LKR 668.7M over 36 months

2.1 Cleaning and Feature Engineering

- **Total ad spends** differed from the sum of the six channels in 16 months by at most LKR 2 (rounding), so it was recalculated from the channels.
- **Outliers** were capped with the IQR rule for EDA. The model uses the original raw data, because the capped values included the real December seasonal peaks.
- **19 features** were added (time, channel mix, engagement, lags, composite indicators). Features built from sales or ad outcomes (ROAS, clicks, conversions) were kept out of the model to avoid leaking the target.

2.2 Exploratory Data Analysis

- **Sales are broadly flat:** LKR 1,029M (2023), 993M (2024) and 1,041M (2025), with peaks around April and December (festive periods).
- **Facebook/Instagram (23.7%) and TV (23.5%)** take the largest share of spend; Radio the smallest (9.0%).
- **Seasonality (r = +0.62) and promotions (+0.58)** relate to sales more strongly than any channel. Among channels, Outdoor (0.43) and Google Search (0.38) are strongest.
- **Radio** shows almost no same-month relationship (0.05) but a two-month lagged correlation of 0.35, a possible delayed effect.
- **Price index and inflation proxy move together** (VIF about 100), a multicollinearity problem addressed in Section 4.

3. Methodology

The model turns each channels spend into a carry-over, diminishing-returns signal, then fits a regularized regression of sales on those signals and the external factors.



Figure 1. *Project workflow from raw data to dashboard (notebooks 01–07)*

1. **Adstock (carry-over).** Part of a month's advertising effect carries into later months: A(t) = X(t) + λ × A(t−1). Decay rates λ: TV 0.4, Facebook/Instagram 0.2, Google Search 0.1, YouTube 0.3, Radio 0.2, Outdoor 0.3.
2. **Hill saturation (diminishing returns).** H(A) = A^s / (A^s + K^s), where K is each channel's median adstocked spend in the training months and s = 1.
3. **Ridge regression with non-negative media effects.** Features are standardised and a ridge penalty keeps coefficients stable with 30 training months. Media coefficients are constrained to be zero or positive, because advertising cannot reduce sales.
4. **Train/test split.** Training: January 2023 – June 2025 (30 months). Test: July – December 2025 (6 months), used only for final evaluation.
5. **Hyperparameters.** The ridge strength (alpha) was chosen by time-series cross-validation (3 folds, no shuffling); the best value was 10.
6. **Inputs.** 13 features: the six transformed channels and seven external factors (inflation proxy removed, Section 4.2).

ROI definitions. A channel's contribution is predicted sales minus predicted sales with that channel switched off in every month. ROI = contribution ÷ spend. Marginal ROI = extra sales from a 1% spend increase ÷ the extra spend; this is the measure that guides budget decisions. Base sales = predicted sales minus all channel contributions.

4. Model Results and Validation

The final model predicts the six unseen test months with 1.13% average error, less than half the error of ordinary regression, and passes every validation check.

4.1 Test Performance

Table 4. *Test-period performance, July – December 2025*

Model	MAE (LKR M)	RMSE (LKR M)	R^2	MAPE	Durbin-Watson
Mean baseline	4.00	6.03	−0.34	4.31%	1.15
OLS regression	2.51	3.22	0.62	2.81%	0.20
Ridge MMM (final)	**0.98**	**1.30**	**0.938**	**1.13%**	**2.22**

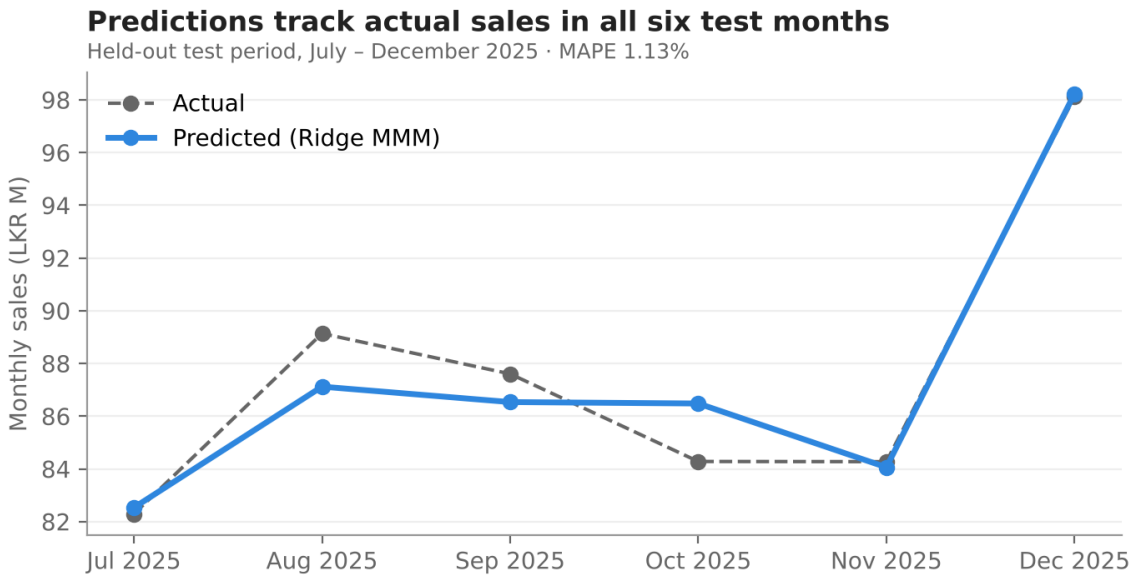


Figure 2. *Actual vs predicted monthly sales in the test period*

The largest monthly miss is October (2.6%); the December festive peak is predicted almost exactly.

4.2 Validation Checks

Table 5. *Model validation checks*

Check	What it tests	Result
Multicollinearity (VIF)	Inputs that move together	Inflation proxy (VIF 111) removed; highest VIF now 6.3; CV error fell from LKR 4.38M to 4.33M
Non-negative media	Implausible negative effects	Radio's negative coefficient fixed; all media effects ≥ 0
Residual diagnostics	Autocorrelation, normality, constant variance	Durbin-Watson 2.18; Ljung-Box, Shapiro-Wilk and Breusch-Pagan all p > 0.05
Bootstrap (1,000 samples)	Uncertainty of channel ROI	Google Search (1.7–6.6) and YouTube (0.5–7.7) clearly positive; other intervals reach zero
Alternative model	Does the ranking depend on our curves?	Log-spend Ridge less accurate (R^2 0.872) but same main ranking (rank correlation 0.87)
Adstock/saturation sensitivity	Do curve settings drive the result?	Google Search and YouTube top two, Radio no effect, in all 9 settings

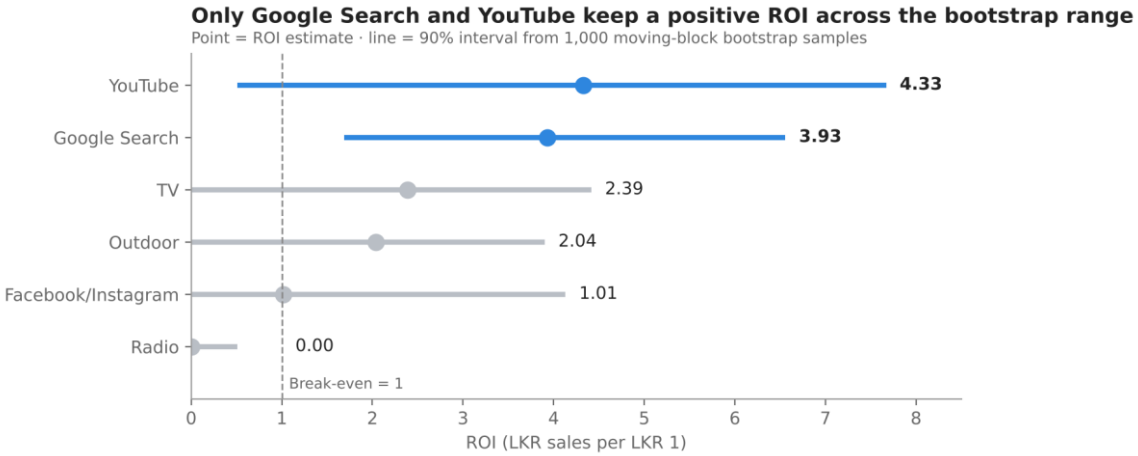


Figure 3. *Channel ROI with 90% bootstrap intervals*

Per-channel tuning of the adstock decay rates was also tested. The "best" values changed with the number of CV folds (Google Search 0.2 with 3 folds, 0.7 with 5) and test MAPE rose to about 1.3%, so the configured values were kept: with 30 training months, tuning over-fits.

5. Channel Effectiveness and ROI

YouTube and Google Search bring in the most sales per rupee and still have room to grow; Facebook/Instagram is past break-even and Radio shows no measurable effect.

5.1 Base Sales vs Advertising

Of LKR 3,061.8M predicted sales over 36 months, **LKR 1,520.1M (49.6%) is base sales** that would happen without advertising, and **LKR 1,541.7M (50.4%) is driven by advertising**. Against LKR 668.7M of ad spend, advertising as a whole returned about **LKR 2.31 per LKR 1**.

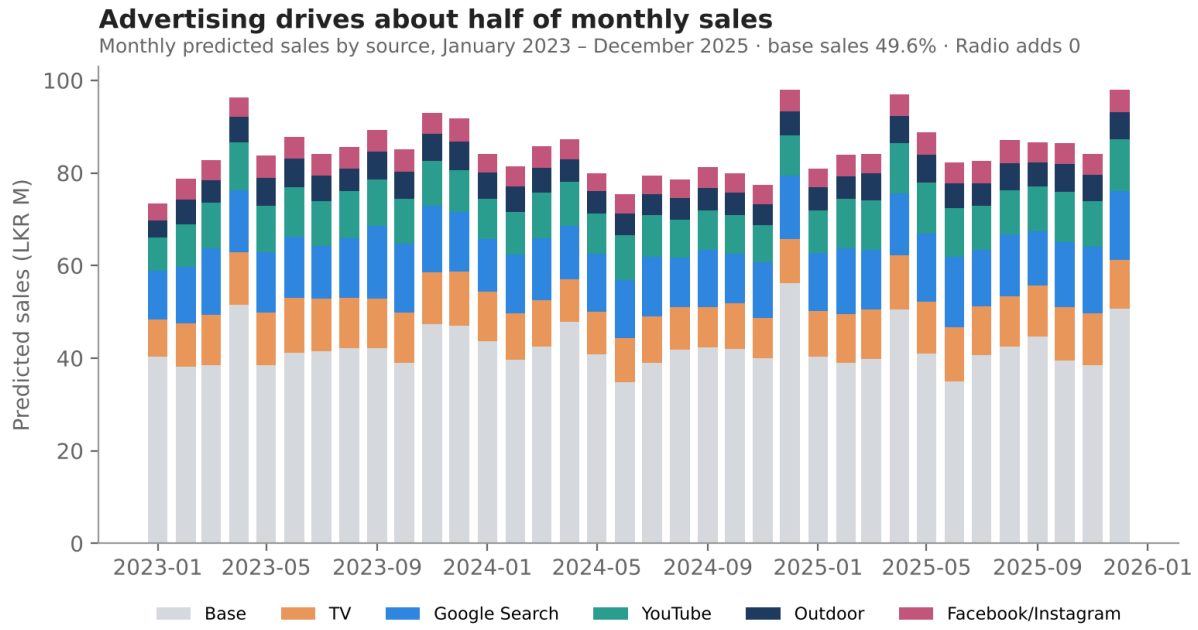


Figure 4. *Monthly predicted sales by source*

5.2 ROI by Channel

Table 6. *Channel ROI, January 2023 – December 2025*

Channel	Spend (LKR M)	Sales caused (LKR M)	ROI	Share of spend	Share of ad sales	Assessment
YouTube	79.9	345.8	4.33	11.9%	22.4%	Under-funded
Google Search	119.4	468.6	3.93	17.8%	30.4%	Under-funded
TV	157.0	375.2	2.39	23.5%	24.3%	Balanced
Outdoor	94.1	191.7	2.04	14.1%	12.4%	Slightly over-funded
Facebook/Instagram	158.3	160.4	1.01	23.7%	10.4%	Over-funded
Radio	60.1	0.0	0.00	9.0%	0.0%	No measurable effect

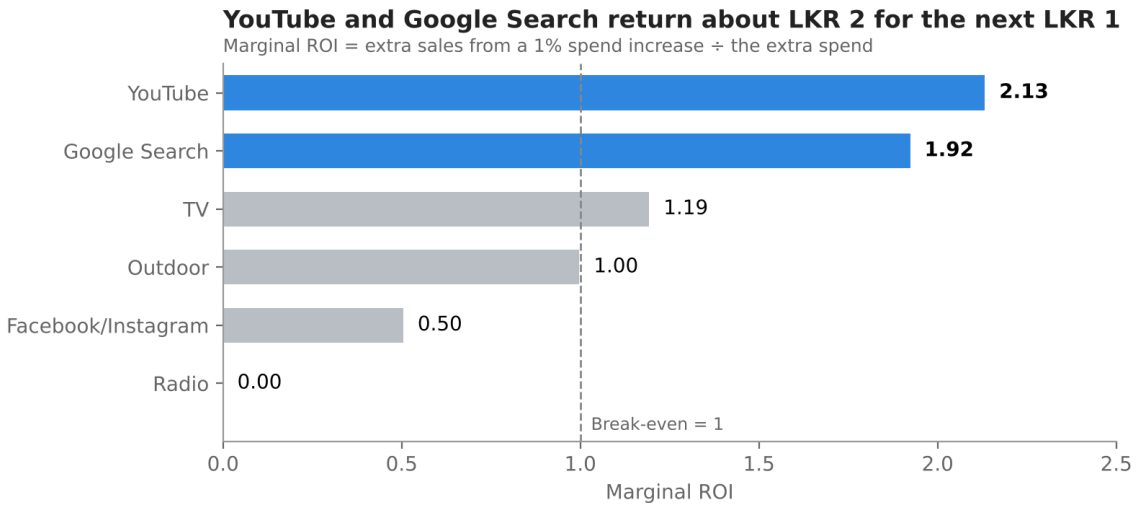


Figure 5. *Marginal ROI by channel*

ROI is the average return on all spend so far; marginal ROI is the return on the next rupee and is what the budget decision rests on. Facebook/Instagram returns only LKR 0.50 for its next rupee.

5.3 Diminishing Returns

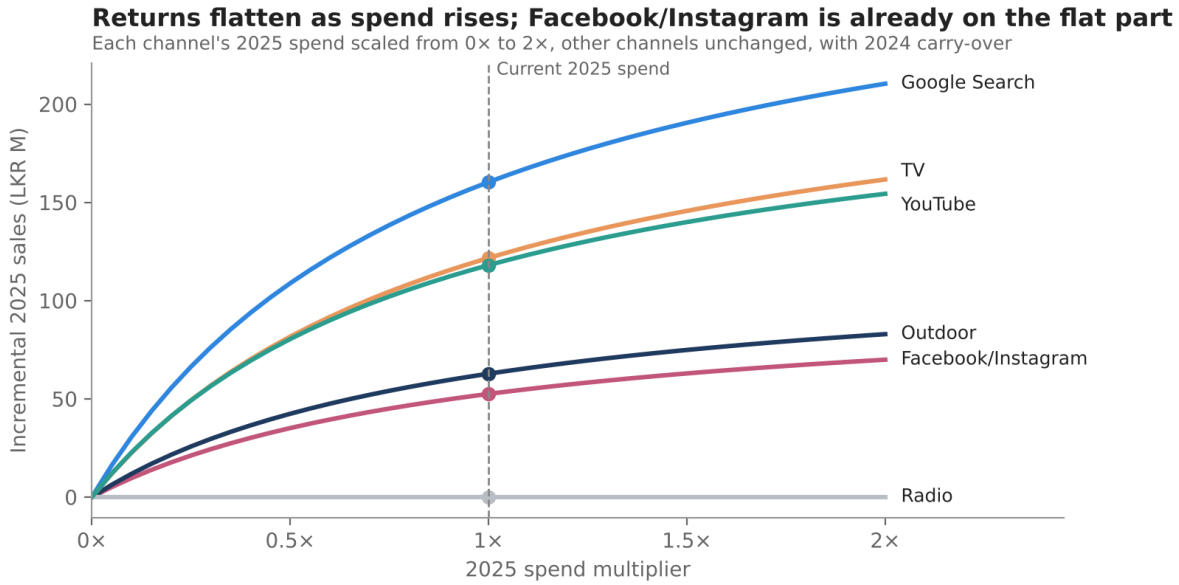


Figure 6. *Response curves for 2025*

Table 7. *Break-even spend levels*

Channel	2025 spend (LKR M)	Spend where marginal ROI falls below 1 (LKR M)	Status
Google Search	43.2	71.3	Room to grow
YouTube	30.6	50.6	Room to grow
TV	56.5	59.4	Some room to grow
Outdoor	33.4	30.1	At break-even
Facebook/Instagram	54.5	21.8	Well past break-even
Radio	22.0	–	No measurable effect

5.4 External Factors

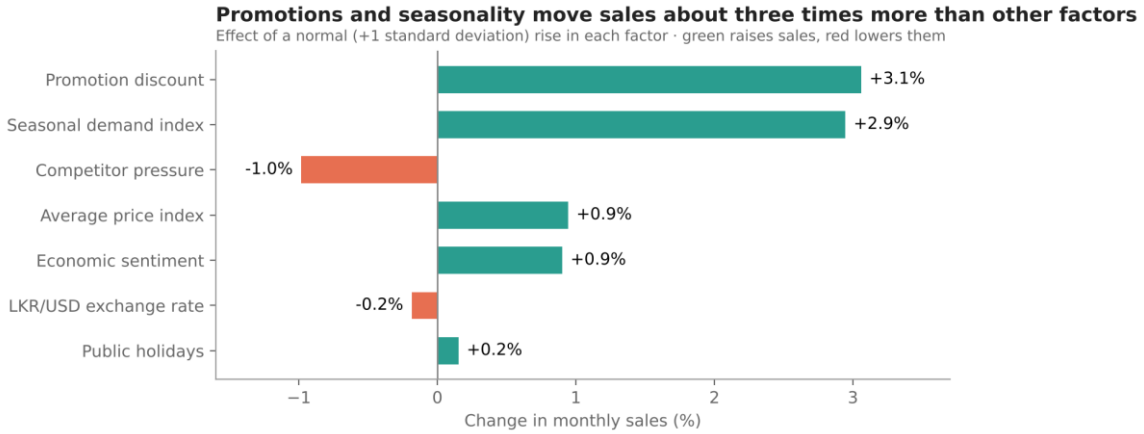


Figure 7. *Effect of external factors on monthly sales*

The December festive season adds about LKR 5.3M a month, and a 5-point higher promotion discount adds about LKR 2.2M (+2.5%). The positive price effect is unusual: the price index rises steadily over time (correlation 0.99 with the month), so it mainly acts as a trend term, not proof that higher prices raise sales.

6. Budget Optimization

With the same LKR 240.3M budget, the optimized split raises predicted 2025 sales from LKR 1,041.5M to **LKR 1,067.4M (+25.9M, +2.5%)**.

6.1 Set-up

The optimizer (SciPy SLSQP) chooses one multiplier per channel on its 2025 monthly spend, keeping each channel's month-to-month pattern. It maximises total predicted 2025 sales from the MMM, including carry-over from 2024. Constraints: total spend equals the 2025 budget, and each channel stays within 70%–130% of its current spend. The result beat all 2,000 random plans that follow the same rules (best random plan LKR 1,060.3M).

6.2 Recommended Split

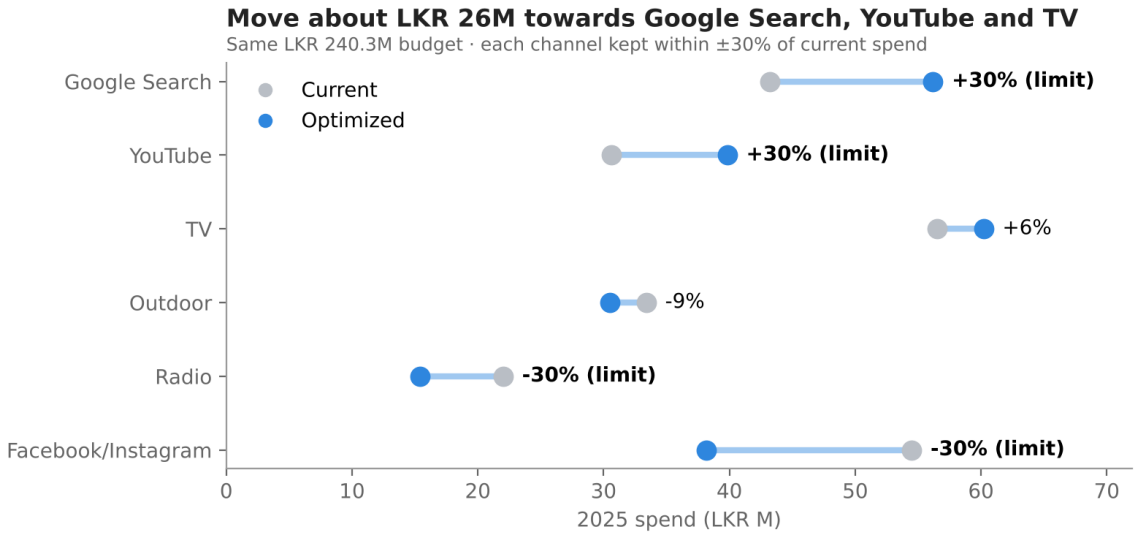


Figure 8. *Current vs optimized 2025 spend by channel*

Table 8. *Recommended budget split (same total budget)*

Channel	Current (LKR M)	Optimized (LKR M)	Change	Optimized share
Google Search	43.2	56.2	+30% (limit)	23.4%
YouTube	30.6	39.8	+30% (limit)	16.6%
TV	56.5	60.2	+6%	25.1%
Outdoor	33.4	30.5	−9%	12.7%
Radio	22.0	15.4	−30% (limit)	6.4%
Facebook/Instagram	54.5	38.1	−30% (limit)	15.9%
Total	**240.3**	**240.3**	**0%**	**100%**

6.3 Budget Scenarios

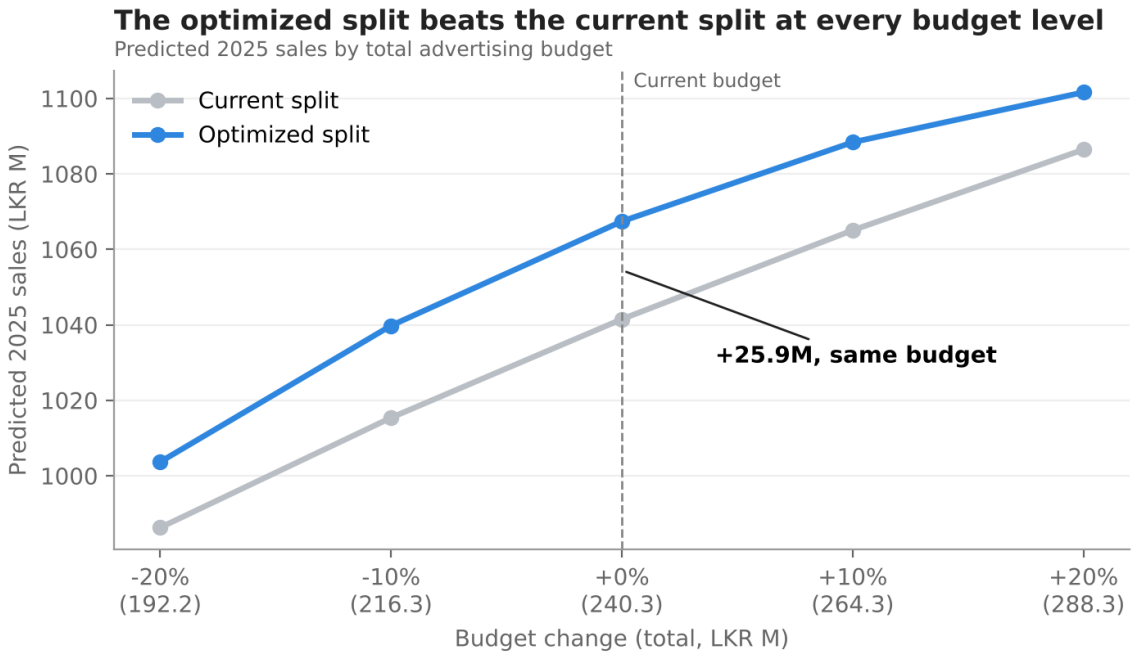


Figure 9. *Predicted 2025 sales across budget scenarios*

A 10% smaller budget with the optimized split gives about the same sales as today's full budget (LKR 1,039.8M vs 1,041.5M), a saving of about LKR 24M. The gain from a better split shrinks above the current budget as channels saturate.

6.4 Sensitivity

Channel limits. Re-running with limits from ±10% to ±50% changes the size of the uplift (LKR 10.3M at ±10% to 36.3M at ±50%) but not its direction: Google Search and YouTube always rise to the limit, and Facebook/Instagram and Radio always fall to it.

Model uncertainty. The model was re-fitted 100 times on resampled data and the budget re-optimized each time.

Table 9. *Re-optimization across 100 bootstrap models*

Channel	Final plan	Samples where increased	Samples where reduced
Google Search	+30%	93%	6%
YouTube	+30%	86%	13%
TV	+6%	57%	41%
Outdoor	−9%	24%	75%
Facebook/Instagram	−30%	29%	69%
Radio	−30%	3%	97%

The uplift was positive in every sample (90% range LKR 14.7M – 63.1M). The shift towards Google Search and YouTube and away from Radio is robust; the TV change is uncertain.

7. Interactive Dashboard

The Power BI dashboard lets a business user explore the results without the notebooks. It reads the CSV outputs in outputs/tables/, so a refresh after re-running the notebooks updates every page.

Table 10. *Dashboard pages*

Page	What it shows
Marketing Mix Performance Overview	Sales, spend and units KPIs; monthly spend vs sales; sales trend
Marketing Channel Performance	Spend split by channel and over time; ROI and marginal ROI; base vs channel contributions
Diminishing Returns	Response curves and marginal ROI by spend, with a channel slicer
Budget Optimization	Current vs optimized budget; reallocation table; sales across budget scenarios
Model Performance & Insights	Model KPIs; actual vs predicted; model comparison; external factor effects

The headline ROI card shows the model-based return (LKR 2.31 per LKR 1), not total revenue ÷ spend (about 4.58), which would count base sales as if advertising caused them. Screenshots of each page are in dashboard/screenshots/.

8. Recommendations, Limitations and Conclusion

8.1 Recommendations

1. **Shift budget from Facebook/Instagram and Radio to Google Search and YouTube,** following the optimized split (+LKR 25.9M expected sales, no extra spend).
2. **Make the changes gradually,** within ±30% per channel per year, and track monthly sales against the model's predictions.
3. **Improve the split before increasing the budget.** A better split is worth about as much as a 10% bigger budget with today's split.
4. **If the budget grows,** put the extra money into Google Search and YouTube first, then TV.
5. **Test Radio before cutting it further,** for example with a regional on/off test; the EDA hinted at a two-month delayed effect.
6. **Plan promotions and seasonal campaigns deliberately,** since they move monthly sales more than any single channel.
7. **Re-run the model and optimization every year** with new data.

8.2 Limitations

- **Small sample:** 36 months (30 for training). Exact ROI values for TV, Outdoor and Facebook/Instagram are uncertain; the rankings are more reliable than the exact numbers.
- **Observational data:** the model shows association, not proven cause and effect. Controlled tests would confirm channel effects.
- **Fixed curve settings:** adstock decay rates are set, not estimated, because tuning over-fits with this sample. The main conclusions hold across nine settings.
- **Revenue, not profit:** margins were not available, so ROI is measured in sales revenue.
- **Known 2025 conditions:** the plan re-optimizes 2025 with its actual seasonality and economy; 2026 planning needs forecasts of these factors.
- **Synthetic data:** the results demonstrate the method rather than a real brand's performance.

8.3 Future Work

- Bayesian MMM (for example PyMC-Marketing) to estimate adstock and saturation with uncertainty from the data.
- Weekly or regional data to increase the sample size.
- Geo or on/off experiments to calibrate channel effects, starting with Radio and Facebook/Instagram.
- Profit-based optimization once margins are available.

8.4 Conclusion

The project delivered a validated Marketing Mix Model, channel ROI estimates, a budget optimizer and an interactive dashboard. The evidence points consistently in one direction: **move budget towards Google Search and YouTube and away from Facebook/Instagram and Radio.** This is expected to raise sales by about 2.5% with the same budget, and the conclusion holds across every robustness check performed.

Appendix: Repository and Reproduction

All code, outputs and stage reports are in github.com/IBL11/mmm-budget-optimization. Running notebooks 01 to 07 in order reproduces every table and figure; 19 unit tests check the evaluation and prediction code.

Table 11. *Notebooks and outputs*

Notebook	Purpose	Main outputs
01_data_cleaning	Quality checks, cleaning	cleaned_marketing_mix_data.csv
02_external_features	Feature engineering	feature_engineered_marketing_mix_data.csv
03_eda	Exploratory analysis, VIF	EDA figures, vif_table.csv
04_transformations	Adstock and saturation exploration	–
05_modelling	Training, evaluation, validation, sensitivity	model files, model_metrics.csv, model_vif.csv, roi_bootstrap_intervals.csv
06_roi_analysis	Contributions, ROI, response curves, external factors	roi_table.csv, contributions.csv, response_curves.csv, external_impact.csv
07_budget_optimisation	Optimization, scenarios, sensitivity	optimal_allocation.csv, budget_scenarios.csv, sensitivity tables

How to reproduce:

1. Clone the repository and create a virtual environment (python -m venv venv), then run pip install -r requirements.txt.
2. Run notebooks 01 → 07 in order (Restart → Run All).
3. Run the tests with python -m pytest -q.
4. Open the dashboard in Power BI Desktop and click Refresh.